

Visual Causal Chain Bookmarking: Causal Credit Selection for Long-Horizon Agentic Reinforcement Learning

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Abstract

Long-horizon RL with sparse terminal rewards must distribute a single scalar reward across H actions, most of which have no causal bearing on the outcome. Existing credit assignment methods (RUDDER, GAE, LOOP, GRPO) redistribute credit temporally or uniformly but never ask *which* timesteps actually caused the outcome. We introduce **Visual Causal Chain Bookmarking (VCBM)**, which detects timesteps where an agent’s action produced a persistent environment-state change, retrieves the most relevant such events via embedding similarity to the terminal state, and restricts policy gradient updates to this subset. We prove the resulting estimator is unbiased and strictly reduces gradient variance relative to uniform credit assignment. We further introduce an HTTP request safety tracker giving a deterministic bound on rollout collection time, eliminating a measured $\sim 14\%$ per-iteration baseline failure rate, and report wall-clock evidence that a naive causal-detector implementation incurs a 1596% rollout-collection overhead — almost entirely (98.0%) attributable to state-snapshot diffing — for which we specify a zero-extra-call, provably equivalent fix.

1. Problem

We extend the standard MDP to an **Agentic Interactive MDP** $\mathcal{M}_{\mathcal{E}} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, R, \gamma, H, \mathcal{E})$, where $\mathcal{A}$ is natural language containing executable code and $\mathcal{E}$ has execution time $\Omega(a, \mathcal{E})$ with $\sup \Omega = \infty$ (non-terminating code is possible). Reward is sparse terminal: $R(s_t, a_t, s_{t+1}) = r_T \in [0, 1]$ at $t=H$, else 0.

Def. (Causally Inert Timestep). For state-change function $f : \mathcal{S} \rightarrow \mathbb{R}^d$ and threshold δ , t is inert if $\|f(s_t) - f(s_{t-1})\|_2 \leq \delta$. Active set $\mathcal{C}(\tau)$, inert set $\mathcal{I}(\tau)$, and $\text{CCR}(\tau) = |\mathcal{C}(\tau)|/H$.

Theorem 1 (Gradient Waste). For $\hat{g} = \sum_t A_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$ with $\mathbb{E}[A_t] = 0$ on $\mathcal{I}(\tau)$: $\text{Var}[\hat{g}] \geq \text{Var}[\sum_{t \in \mathcal{C}(\tau)} A_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]$, equality iff $A_t = 0$ a.s. on $\mathcal{I}(\tau)$. *Proof:* decompose $\hat{g} = \hat{g}_{\mathcal{C}} + \hat{g}_{\mathcal{I}}$; zero-mean inert-step terms add strictly positive variance unless degenerate. $\square$ **Corollary:** LOOP and GRPO incur avoidable variance whenever $\text{CCR}(\tau) < 1$.

2. VCBM Mechanism

Five components: state encoder g_{ϕ} , causal detector $\varphi_{\delta}(s_t, s_{t-1}) = \mathbf{1}[\|f(s_t) - f(s_{t-1})\|_2 > \delta]$, episodic memory $\{k_i = g_{\phi}(s_{t_i}), v_i = (a_{t_i}, G_{t_i})\}$ keyed on detected events, retrieval by cosine similarity ρ_i to terminal state z_H with top- k set $\mathcal{R}_k(\tau)$, and softmax credit weights $w_t \propto \exp(\rho_i / \tau_{\text{temp}})$ on $\mathcal{R}_k(\tau)$, zero elsewhere. The gradient is $\nabla_{\theta} J_{\text{VCC}} = \mathbb{E}_{\tau}[\sum_{t \in \mathcal{R}_k(\tau)} w_t A_{\text{VCC}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]$, replacing LOOP’s uniformly-broadcast leave-one-out scalar $A_{\text{LOOP}}(\tau_i) = R_i - \frac{1}{N-1} \sum_{j \neq i} R_j$ with a sparse, relevance-weighted assignment. Final advantage blends $A_{\text{blend}} = (1-\alpha)A_{\text{LOOP}} + \alpha w_t A_{\text{VCC}}$, $\alpha=0$ recovering standard LOOP exactly.

Theorem 2 (Unbiasedness). If the detector has zero false-negative rate, $\mathbb{E}_{\tau}[\hat{g}_{\text{VCC}}] = \nabla_{\theta} J(\theta)$, since w_t is a θ -independent post-hoc function and zero-weighted terms contribute exactly zero by the policy gradient theorem.

Theorem 3 (Variance Reduction). $\text{Var}[\hat{g}_{\text{VCC}}] \leq \text{Var}[\hat{g}_{\text{LOOP}}] - \sum_{t \in \mathcal{I}(\tau)} \text{Var}[A_t^{\text{LOOP}} \nabla_{\theta} \log \pi_{\theta}(a_t | a_t)]$, immediate from Theorem 1 since VCC zeros all inert-step weights. **Corollary:** $N_{\text{VCC}} \leq N_{\text{LOOP}} \cdot \text{CCR}(\tau)$ in the zero-error limit.

3. HTTP Safety Tracker & Cost of Detection

A tracker $\mathcal{H}$ enforces per-request timeout T_{max} , returning (TIMEOUT, $-\epsilon, s_t$) if $\Omega(a_t, \mathcal{E}) > T_{\text{max}}$. **Theorem 4:** $\Omega_{\mathcal{H}}(\tau) \leq H(T_{\text{max}} + T_{\text{gen}}) + T_{\text{overhead}}$ by induction over turns, giving $P_{\text{collapse}} = 0$ — confirmed across 200 iterations with zero crashes, against a measured 14% baseline collapse rate.

The naive detector implementation reads a full state snapshot before/after every turn (`get_full_state_snapshot()`), doubling HTTP calls per turn. The one observed collapse incident was a malformed retry loop:

```
while True: try: x.extend(api.top_tracks()) except Exception: if e.code==403: break
```

the `e.code` attribute does not exist on the raised exception, so the generic `except` swallows the `ReadTimeout` and retries identically forever; this blocked for 243s against a 100s per-call timeout, requiring a forced SIGKILL restart. $\mathcal{H}$ must wrap the *individual* HTTP call, not the code block, to bound this case. Measured cost over a matched 50-step window (Qwen2.5-1.5B-Instruct, LoRA rank 8, single RTX 4080):

Phase	LOOP (s)	VCBM-naive (s)	Rel. cost
Model setup / log-probs / optim. / grad. steps	143	125	−13% avg
Rollout collection	88.5	1500	+1596%
Compute gradients / adv. filtering	73.6	62.1	−16% avg
Total	305	1687	+453%

Prop. 1: rollout collection accounts for 98.0% of the total cost differential, isolating state-diffing as the dominant overhead source. Since AppWorld already tracks each request’s HTTP method and status, a zero-extra-call detector is available: $\varphi^{\text{http}}(t) = \mathbf{1}[\exists \text{req} : \text{method} \in \{\text{POST}, \text{PUT}, \text{DELETE}, \text{PATCH}\} \wedge \text{status} < 400]$. **Prop. 2 (Equivalence):** under REST conformance, $\varphi^{\text{http}}(t) = \varphi_\delta(s_t, s_{t-1})$ for all t , so this is a drop-in replacement with no change to Theorems 2–3; projected speedup $\sim 16\text{--}17\times$ on the dominant cost term, not yet re-measured at full scale.

4. Four-Factor Validation Protocol (Proposed)

Define episodes-to-threshold $N_\theta(\text{method}) = \min\{N : \frac{1}{100} \sum_{i=N-99}^N \mathbf{1}[\text{success}_i] \geq \theta\}$, $\theta=0.90$. **(F1) Convergence speed:** VCBM vs. RUDDER, 5 seeds, MiniGrid-KeyCorridorS3, report $N_{0.90}^{\text{RUDDER}}/N_{0.90}^{\text{VCBM}}$. **(F2) Horizon scaling:** repeat F1 across S3–S6 ($H \approx 30$ to 96) with $\text{CCR}(\tau)$ decreasing in H ; define $\text{Gap}(H) = N_{0.90}^{\text{RUDDER}}(H) - N_{0.90}^{\text{VCBM}}(H)$ and test whether $\text{Gap}(H)$ increases significantly via linear regression. **(F3) Revisit robustness:** a gridworld requiring $\ell \in \{1, 3, 5\}$ revisits before the productive transition; VCBM vs. plain leave-one-out, $\text{LoopGap}(\ell) = N_{0.90}^{\text{LOOP}}(\ell) - N_{0.90}^{\text{VCBM}}(\ell)$, plus a qualitative *Loop Disambiguation Rate* (fraction of successful episodes where the retrieved chain identifies the productive revisit). **(F4) Significance:** ≥ 5 seeds throughout; Welch’s t -test with Cohen’s d for all $N_{0.90}$ comparisons; regression slopes for $\text{Gap}(H)/\text{LoopGap}(\ell)$ tested against zero; Bonferroni correction within each factor; no single-seed result reported as a finding.

Ablation design. The *LOOP-matched* condition (identical context length, rollout count, GPU allocation, but $\alpha=0$) isolates the causal mechanism from configuration effects; *random-anchors* tests whether specific causal identification matters versus mere sparsification.

Condition	α	Detector	HTTP Tracker
LOOP-original / LOOP-matched	0.0	—	✗
VCBM-no-tracker	0.1	✓	✗
VCBM-random-anchors	0.1	random	✓
VCBM-uniform-credit	0.1	✓ (uniform w_t)	✓
VCBM-full	0.1	✓	✓

5. Empirical Evidence & Status

Two independent 200-iteration runs (Qwen2.5-1.5B-Instruct, LoRA rank 8, 9.23M/1.55B trainable params, AdamW 5×10^{-5} , PPO $\epsilon=0.1$, $\alpha=0.1$, $\text{top-}k=5$, $\tau_{\text{temp}}=0.1$) confirm Theorem 4 directly: zero crashes in either configuration (4096-token eager-mode, 4 rollouts/iter; and 8192-token CUDA-graph-enabled, 8 rollouts/iter), no high-KL policy collapse ($\text{all_kl_estimate} \in [-0.86, 0.78]$), and advantage-filtering fractions of 0.75–0.88 indicating room for sample efficiency gains via Corollary 1. The compiled, larger-cache configuration reduced rollout cost from 36.9 to 14.15s/iteration despite collecting $2\times$ the rollouts, attributable to CUDA graphs and KV-cache sizing rather than the detector itself. These results substantiate the safety claim (Theorem 4) and the cost-attribution claim (Prop. 1); the variance-reduction claim (Theorem 3) awaits the LOOP-matched ablation and four-factor protocol above, both specified but not yet run, alongside re-measurement under the corrected detector of Section 3.

References: Arjona-Medina et al., *RUDDER*, NeurIPS 2019 · Schulman et al., *GAE*, ICLR 2016 · Schulman et al., *PPO*, arXiv:1707.06347 · Apple ML Research, *LOOP*, 2025 · DeepSeek-AI, *GRPO*, 2024 · Andrychowicz et al., *HER*, NeurIPS 2017 · Pritzel et al., *Neural Episodic Control*, ICML 2017 · Trivedi et al., *AppWorld*, ACL 2024 · Qwen Team, *Qwen2.5 Technical Report*, arXiv:2412.15115 · Hu et al., *LoRA*, arXiv:2106.09685 · Loshchilov & Hutter, *AdamW*, ICLR 2019 · Kwon et al., *vLLM/PagedAttention*, SOSP 2023 · Zhao et al., *PyTorch FSDP*, VLDB 2023.